

Suggested Projects for Practice

Mini Project Suggestions (1-2 Days)

1. **URL Shortener REST API:** Includes custom dynamic hashing, redirection, and hit analytics.
2. **Contact Book API:** Implement CRUD with pagination, CSV file import/export, and validation.
3. **Task Tracker API:** In-memory task system using role-based Security (USER, ADMIN).

Major Enterprise Project Suggestions (1-2 Weeks)

1. **E-Commerce Microservice API Engine:** Complete order processing, payment gateway mock integration, JWT multi-role auth (CUSTOMER, SELLER, ADMIN), product specifications search engine, and PDF invoice generation.
2. **Healthcare Management System:** Patient record tracking, doctor appointment scheduling using optimistic locking to prevent double bookings, file uploads for lab diagnostics, and audit log mechanisms.

Summary

Congratulations! Over this 3-part series, you have transitioned from basic REST principles to building production-grade enterprise Java backends. You now possess the skills to write secure, scalable, well-documented, and performant Spring Boot RESTful microservices. Happy coding!
